Chapter 14: Maps and Locations

Worksheet 14: Maps and Locations

Learning Objective: Read simple maps, use cardinal directions (N, S, E, W), and use scale to estimate real distances from a map.

Worked Example:
On a map, the scale is 1 cm = 5 km. If the distance between two towns on the map is 6 cm, what is the actual distance?

Solution: Actual distance $= 6 \times 5 = 30$ km.

Questions (1–20)

1. Name the four main (cardinal) directions.

2. If you are facing North and turn to face East, did you turn clockwise or anticlockwise?

3. On a map, which direction is usually shown at the top?

4. If a place is to the right of North on a map, which direction is it?

5. What is the opposite direction of South?

6. On a map with scale 1 cm = 10 km, a distance of 4 cm represents how many actual km?

7. A school is North of a park, and a market is South of the same park. If you walk from the school to the market, in what direction do you travel?

8. On a map, a river runs North to South. In what general direction would a bridge crossing it most likely run?

9. A treasure map shows: walk 3 cm East, then 2 cm North. If scale is 1 cm = 2 km, find the actual eastward and northward distances.

10. On a map, the distance between two cities is 8 cm with scale 1 cm = 25 km. Find the actual distance.

11. On a map (scale 1 cm = 1 km), the school is 2 cm East and 1 cm North of your house, and the park is 1 cm West and 3 cm North of your house. Describe, using directions and distances, how to walk from school to your house.

12. Point B is 5 cm from Point A on a map with scale 1 cm = 4 km. A delivery person travels at 20 km/h. How long will the journey take?

13. On a treasure hunt map with scale 1 cm = 50 m, a path measures 7 cm. Find the actual length of the path in metres.

14. Two towns are 15 km apart in real life. If the map scale is 1 cm = 3 km, how many cm apart will they be shown on the map?

15. A map shows a straight road running exactly East to West. If you start walking from the West end, in which direction are you walking?

16. A treasure map (scale 1 cm = 2 km) shows: walk 4 cm North, then 3 cm East, then 2 cm South. Find the net distance travelled North and the net distance travelled East, in actual km.

17. A delivery route (scale 1 cm = 5 km) goes 6 cm East, then 4 cm North, then 6 cm West back to a point directly North of the depot. Find the actual distance of each leg and the total distance travelled.

18. A flight path of 1,100 km is shown on a map with scale 1 cm = 200 km. How many cm would this distance be on the map?

19. Describe a simple map of your route from home to school, marking at least 2 landmarks and the main direction of travel (N/S/E/W).

20. If you wanted to use a map app to find the distance from your home to a relative's house in another city, what information would you need to estimate the real distance?

Reflection Question: Why do you think maps always show a direction (North) and a scale? What problem would happen if a map did not have these?

Real-Life Application Question: Look at any map (school atlas, neighbourhood map, or a map app) and identify the scale and at least one direction shown on it.